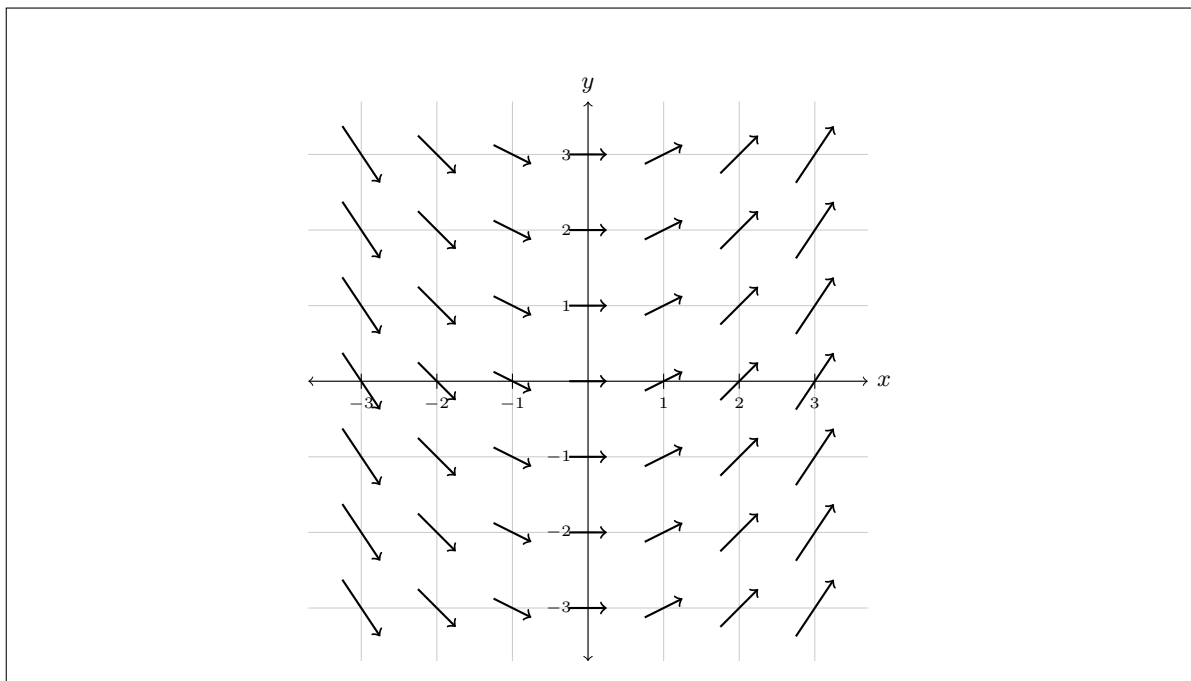


**Topics:** 16.1. Vector Fields; 16.2. Line Integrals (First Part)

Please put away all electronic devices, including cell phones and calculators.

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1. Sketch the vector field  $\mathbf{F} = 2\mathbf{i} + x\mathbf{j}$ .



2. Find the gradient vector field  $\nabla f$  of  $f$ .

(a)  $f(x, y) = 2y \sin(xy)$

Since  $f(x, y) = 2y \sin(xy)$ , we have

$$\begin{aligned}\nabla f(x, y) &= f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j} \\ &= (2y \cos(xy) \cdot y)\mathbf{i} + (2y \cdot x \cos(xy) + 2 \sin(xy) \cdot 1)\mathbf{j} \\ &= 2y^2 \cos(xy)\mathbf{i} + (2xy \cos(xy) + 2 \sin(xy))\mathbf{j}\end{aligned}$$

(b)  $f(x, y, z) = x^3 y e^{y/z}$

Since  $f(x, y, z) = x^3 y e^{y/z}$ , we have

$$\begin{aligned}\nabla f(x, y, z) &= f_x(x, y, z)\mathbf{i} + f_y(x, y, z)\mathbf{j} + f_z(x, y, z)\mathbf{k} \\ &= 3x^2 y e^{y/z}\mathbf{i} + x^3 \left( y e^{y/z} \frac{1}{z} + e^{y/z} \cdot 1 \right)\mathbf{j} + \left( x^3 y e^{y/z} \left( -\frac{y}{z^2} \right) \right)\mathbf{k} \\ &= 3x^2 y e^{y/z}\mathbf{i} + x^3 e^{y/z} \left( \frac{y}{z} + 1 \right)\mathbf{j} - \frac{x^3 y^2}{z^2} e^{y/z}\mathbf{k}\end{aligned}$$

3. Determine whether the vector field  $\mathbf{F}$  is a gradient vector field. If it is, find a function  $f$  such that  $\mathbf{F} = \nabla f$ . If it is not, explain why using Clairaut's Theorem.

(a)  $\mathbf{F} = 4x^3y^3 \mathbf{i} + (3x^4y^2 - 4y^3) \mathbf{j}$

We will find a function  $f$  such that  $\mathbf{F} = \nabla f$  by taking antiderivatives:

$$f(x, y) = \int 4x^3y^3 dx = x^4y^3 + g_1(y)$$

for some function  $g_1(y)$  and

$$f(x, y) = \int (3x^4y^2 - 4y^3) dy = x^4y^3 - y^4 + g_2(x)$$

for some function  $g_2(x)$ . Therefore, the function

$$f(x, y) = x^4y^3 - y^4$$

has the property that  $\mathbf{F} = \nabla f$ . Indeed, we verify that

$$\nabla(x^4y^3 - y^4) = 4x^3y^3 \mathbf{i} + (3x^4y^2 - 4y^3) \mathbf{j}.$$

(b)  $\mathbf{F} = x \cos y \mathbf{i} + x \sin y \mathbf{j}$

Suppose that  $\mathbf{F}$  is a gradient field, so that  $\mathbf{F} = \langle f_x, f_y \rangle$ . Differentiating, we find that

$$f_{xy} = -x \sin y \quad \text{and} \quad f_{yx} = \sin y$$

We see that  $f_{xy} \neq f_{yx}$ , which contradicts Clairaut's theorem. Thus, no such function  $f$  can exist and  $\mathbf{F}$  is not a gradient field.

4. Evaluate the line integral  $\int_C f(x, y) ds$  where  $f$  and  $C$  are given.

(a)  $f(x, y) = \frac{x}{y}$ , where  $C$  is given by  $x = t^3$ ,  $y = t^4$ ,  $1 \leq t \leq 2$ .

The curve  $C$  is parameterized by the vector function  $\mathbf{r}(t) = \langle t^3, t^4 \rangle$  for  $1 \leq t \leq 2$ . Then

$$\mathbf{r}'(t) = \langle 3t^2, 4t^3 \rangle$$

and hence

$$|\mathbf{r}'(t)| = \sqrt{(3t^2)^2 + (4t^3)^2} = \sqrt{9t^4 + 16t^6} = t^2 \sqrt{9 + 16t^2}.$$

We now compute the line integral:

$$\int_C \frac{x}{y} ds = \int_1^2 \frac{t^3}{t^4} |\mathbf{r}'(t)| dt = \int_1^2 \frac{1}{t} \cdot t^2 \sqrt{9 + 16t^2} dt = \int_1^2 t \sqrt{9 + 16t^2} dt.$$

Let  $u = 9 + 16t^2$ , then  $du = 32t dt$ . When  $t = 1$ ,  $u = 25$ , and when  $t = 2$ ,  $u = 73$ . Thus

$$\begin{aligned}\int_1^2 t\sqrt{9+16t^2} dt &= \frac{1}{32} \int_{25}^{73} \sqrt{u} du \\ &= \frac{1}{32} \int_{25}^{73} u^{1/2} du \\ &= \frac{1}{32} \left( \frac{2}{3} u^{3/2} \right) \Big|_{25}^{73} \\ &= \frac{1}{48} \left( 73^{3/2} - 25^{3/2} \right).\end{aligned}$$

Therefore,

$$\int_C \frac{x}{y} ds = \frac{1}{48} \left( 73^{3/2} - 25^{3/2} \right).$$

(b)  $f(x, y) = x + y$ , where  $C$  is the upper semicircle  $x^2 + y^2 = 4$  and  $y \geq 0$ .

We parameterize  $C$  with the vector function  $\mathbf{r}(t) = \langle 2 \cos t, 2 \sin t \rangle$  for  $0 \leq t \leq \pi$ . Then

$$\mathbf{r}'(t) = \langle -2 \sin t, 2 \cos t \rangle$$

and hence

$$|\mathbf{r}'(t)| = \sqrt{(-2 \sin t)^2 + (2 \cos t)^2} = \sqrt{4(\sin^2 t + \cos^2 t)} = \sqrt{4 \cdot 1} = 2.$$

We now compute the line integral,

$$\begin{aligned}\int_C x + y ds &= \int_0^\pi (2 \cos t + 2 \sin t) \cdot 2 dt \\ &= 4 \int_0^\pi \cos t dt + 4 \int_0^\pi \sin t dt \\ &= 4 \cdot 0 + 4 \cdot 2 \\ &= 8.\end{aligned}$$

(c)  $f(x, y) = xy$ , where  $C$  is the arc of the curve  $y = \sqrt{x}$  from  $(4, 2)$  to  $(9, 3)$ .

We parameterize  $C$  with the vector function  $\mathbf{r}(t) = \langle t^2, t \rangle$  for  $2 \leq t \leq 3$ . Then

$$\mathbf{r}'(t) = \langle 2t, 1 \rangle$$

and hence

$$|\mathbf{r}'(t)| = \sqrt{4t^2 + 1}.$$

We now compute the line integral,

$$\begin{aligned}
 \int_C xy \, ds &= \int_2^3 t^2 \cdot t \cdot \sqrt{4t^2 + 1} \, dt \\
 &= \int_2^3 t^3 \sqrt{4t^2 + 1} \, dt & u = 4t^2 + 1, \, du = 8t \, dt \\
 &= \frac{1}{8} \int_{17}^{37} \frac{u-1}{4} \sqrt{u} \, du \\
 &= \frac{1}{32} \int_{17}^{37} u^{3/2} - u^{1/2} \, du \\
 &= \frac{1}{32} \left( \frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2} \right) \Big|_{17}^{37} \\
 &= \frac{1}{32} \left( \left( \frac{2}{5} \cdot 37^{5/2} - \frac{2}{3} \cdot 37^{3/2} \right) - \left( \frac{2}{5} \cdot 17^{5/2} - \frac{2}{3} \cdot 17^{3/2} \right) \right)
 \end{aligned}$$

5. The **average value** of a function  $f(x, y)$  along a curve  $C$  is defined by

$$\bar{f} = \frac{1}{\text{length}(C)} \int_C f(x, y) \, ds.$$

Find the average value of  $f$  along  $C$ .

(a)  $f(x, y) = \frac{x-1}{1+y^2}$ , where  $C$  is the line segment from  $(1, 0)$  to  $(3, 1)$ .

We begin by calculating  $\text{length}(C)$  using the distance formula,

$$\text{length}(C) = \sqrt{(3-1)^2 + (1-0)^2} = \sqrt{2^2 + 1^2} = \sqrt{5}.$$

Now, we parameterize  $C$  with the vector function  $\mathbf{r}(t) = \langle 2t + 1, t \rangle$  for  $0 \leq t \leq 1$ . Then

$$\mathbf{r}'(t) = \langle 2, 1 \rangle$$

and hence

$$|\mathbf{r}'(t)| = \sqrt{2^2 + 1^2} = \sqrt{5}.$$

Now, we compute the average value by evaluating the line integral,

$$\begin{aligned}
 \bar{f} &= \frac{1}{\text{length}(C)} \int_C \frac{x-1}{1+y^2} \, ds \\
 &= \frac{1}{\sqrt{5}} \int_0^1 \frac{(2t+1)-1}{1+t^2} \cdot \sqrt{5} \, dt \\
 &= \int_0^1 \frac{2t}{1+t^2} \, dt \\
 &= \ln(1+t^2) \Big|_0^1 \\
 &= \ln 2 - \ln 1 \\
 &= \ln 2.
 \end{aligned}$$

(b)  $f(x, y) = xe^y$ , where  $C$  is the right half of the unit circle centered at the origin.

By geometry, we have that

$$\text{length}(C) = \frac{1}{2} \cdot 2\pi = \pi.$$

Now, we parameterize  $C$  with the vector function  $\mathbf{r}(t) = \langle \cos t, \sin t \rangle$  for  $-\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$ . Then

$$\mathbf{r}'(t) = \langle -\sin t, \cos t \rangle$$

and hence

$$|\mathbf{r}'(t)| = \sqrt{(-\sin t)^2 + (\cos t)^2} = \sqrt{\sin^2 t + \cos^2 t} = \sqrt{1} = 1.$$

Now, we compute the average value by evaluating the line integral,

$$\begin{aligned}\bar{f} &= \frac{1}{\text{length}(C)} \int_C xe^y \, ds \\ &= \frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos t \cdot e^{\sin t} \cdot 1 \, dt \\ &= \frac{1}{\pi} e^{\sin t} \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \\ &= \frac{1}{\pi} \left( e^{\sin(\pi/2)} - e^{\sin(-\pi/2)} \right) \\ &= \frac{1}{\pi} (e - e^{-1}).\end{aligned}$$

## SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

**Vector Fields.** Let  $D$  be a region in  $\mathbb{R}^2$ . A **vector field** on  $D$  is a function that assigns to each point  $(x, y)$  in  $D$  a vector  $\mathbf{F}(x, y)$ . A vector field in  $\mathbb{R}^2$  has the form

$$\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}.$$

Similarly, if  $E$  is a region in  $\mathbb{R}^3$ , then a vector field on  $E$  assigns to each point  $(x, y, z)$  in  $E$  a vector  $\mathbf{F}(x, y, z)$ . A vector field in  $\mathbb{R}^3$  has the form

$$\mathbf{F}(x, y, z) = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}.$$

A vector field  $\mathbf{F}$  is called a **gradient field** if there exists a function  $f$  such that

$$\mathbf{F} = \nabla f.$$

In this case,  $\mathbf{F}$  is also said to be **conservative**, and  $f$  is called a **potential function** for  $\mathbf{F}$ .

**Scalar Line Integrals.** Let  $C$  be a smooth curve given by the vector function  $\mathbf{r}(t)$ ,  $a \leq t \leq b$ . If  $f$  is a function whose domain contains  $C$ , then the **scalar line integral of  $f$  along  $C$**  is

$$\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| \, dt.$$

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### Suggested Textbook Problems

Section 16.1: 1-22, 25-34

Section 16.2 (First Part): 1-4, 9-12, 27, 28